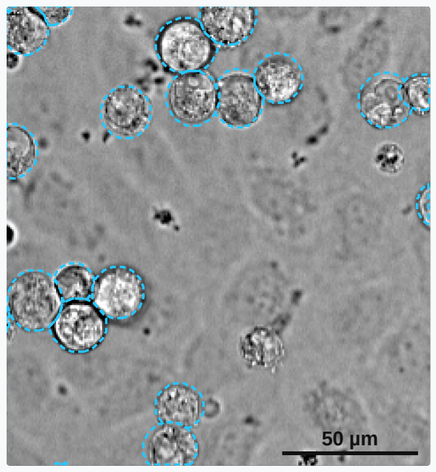
a When a pre-trained model performs poorly, find the mismatch first

field 1 · field 2 (F1 per field) · model · manual annotation

1 Check the normalisation

tile: forgot to normalise

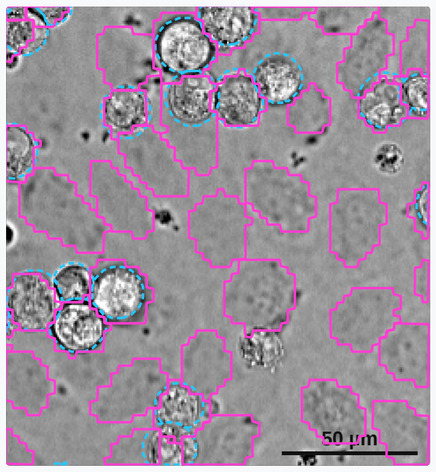


F1 (mean of 2 fields)

forgot to normalise	**0.00**
percentile (default)	0.59
min–max	0.61
histogram equalisation	0.51
CLAHE	0.51

2 Check the object scale

tile: input downsampled 4×

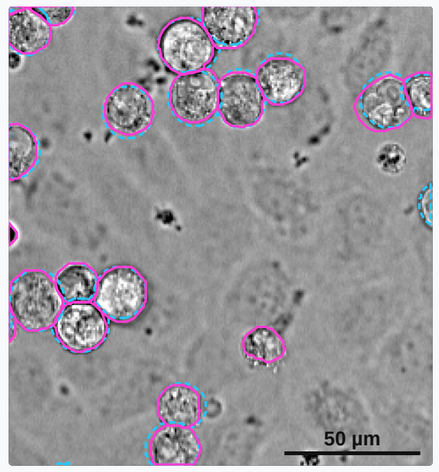


F1 (mean of 2 fields)

downsampled 4×	**0.30**
downsampled 2×	0.41
native (default)	0.59
upsampled 2×	0.56

3 Check the post-processing

tile: low-contrast objects removed



F1 (mean of 2 fields)

none (default)	0.59
stricter probability threshold	0.59
looser flow threshold	0.56
low-contrast objects removed	**0.86**

4 If nothing works, fine-tune or train

b Predicted images: what will you measure?

Used as an intermediate step

visualise · segment · detect · register

→ **Validate the downstream output**

e.g. segment on the predicted image, compare with annotations of acquired images

Used as the measurement itself

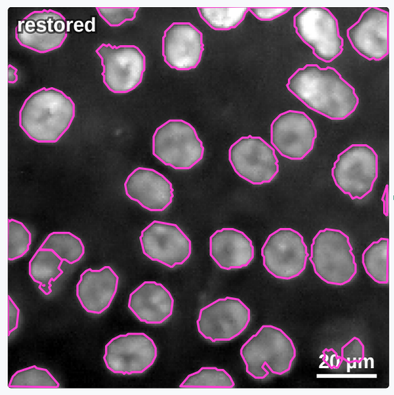
intensity · abundance · localisation

→ **Validate the measurement against acquired data**

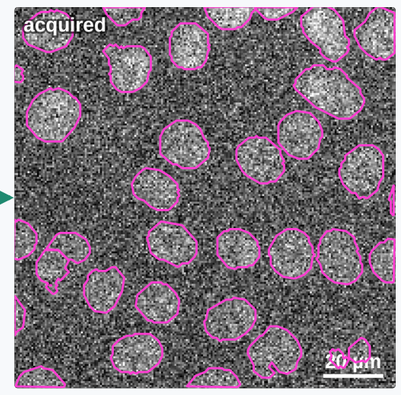
e.g. predicted vs acquired fluorescence, in independent paired samples

Label predicted images as predicted; do not describe predicted fluorescence as measured.

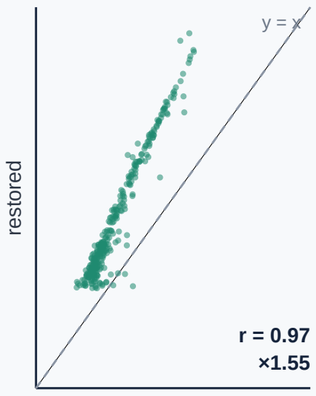
c Denoising: measure intensity in the acquired image



mask from here

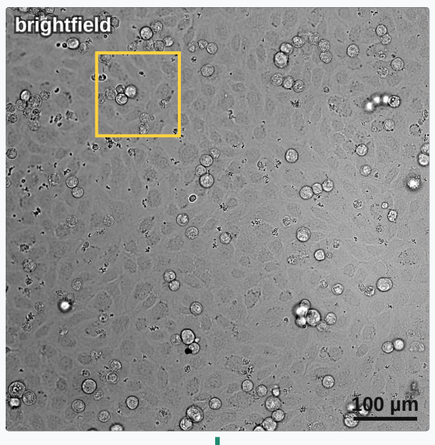


measure here

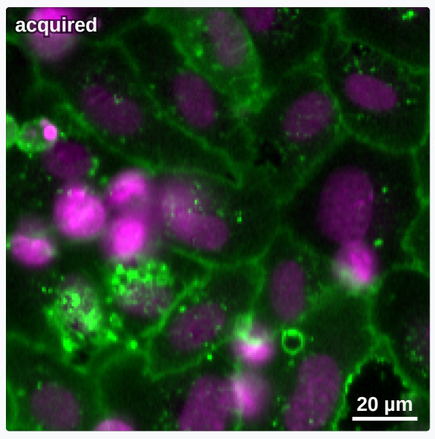


d Artificial labelling: validate what you will measure

model · manual, annotator 1 · manual, annotator 2



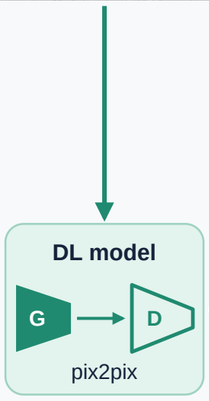
DAPI + PECAM-1



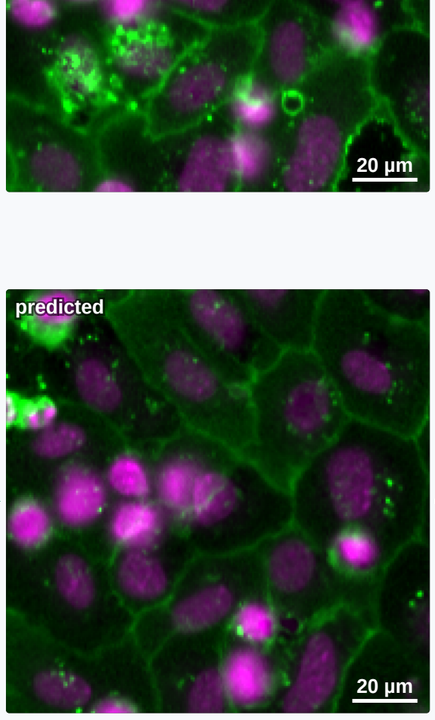
manual annotation

model segmentation

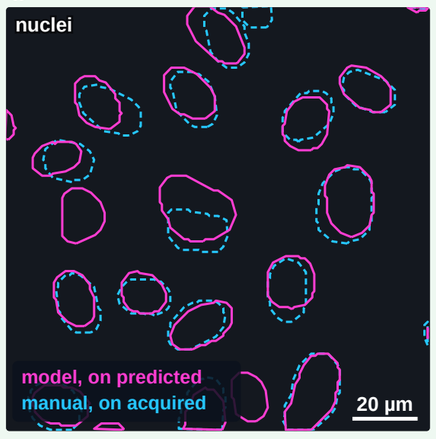
DL model



pix2pix

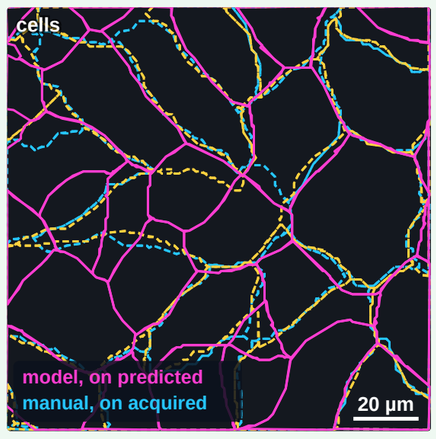


Intermediate step: to aid segmentation



Nuclei · F1

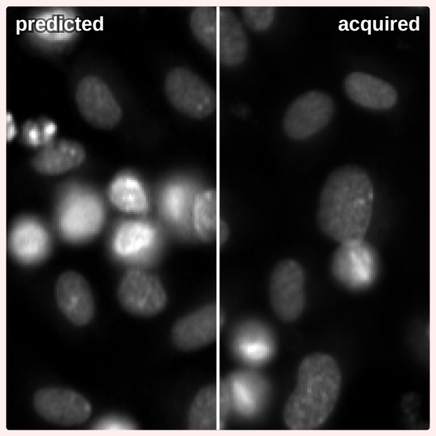
model vs manual nuclei	0.83



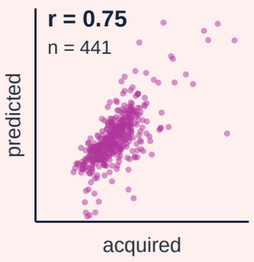
Cells · F1

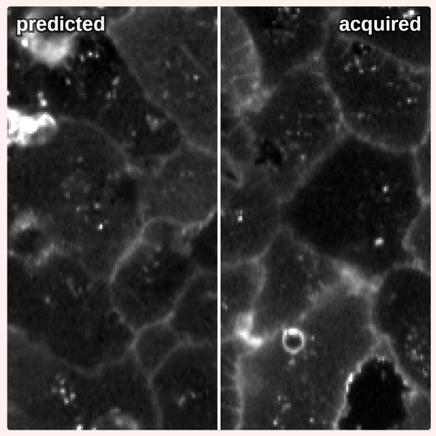
annotator 1 vs 2	0.90
model vs annotator 1	0.78
model vs annotator 2	0.75

Measurement itself: intensity

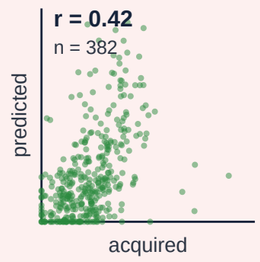


DAPI, per nucleus



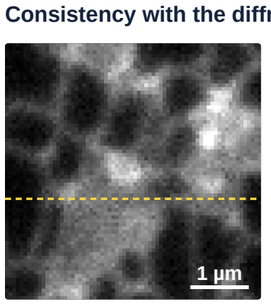


PECAM-1, per cell

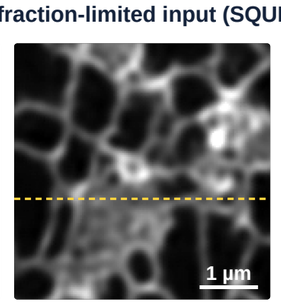


e Super-resolution: check against the input and, where possible, higher-resolution data

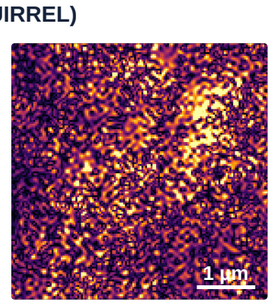
Consistency with the diffraction-limited input (SQUIRREL)



low-res input



predicted



error map

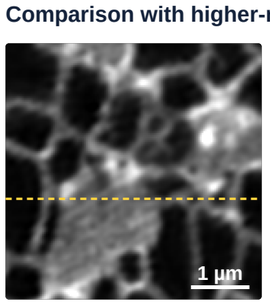
0 · error (a.u.) · high

RSP with the input

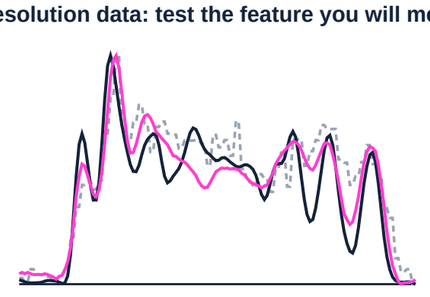
predicted	0.98
higher-res	0.93

0.90 · 0.95 · 1.00

Comparison with higher-resolution data: test the feature you will measure



higher-res



Tubule width (FWHM, nm), 1,301 locations

low-res input	293
predicted	199
higher-res	212

100 · 200 · 300 · 400

predicted − higher−res, same location: −9 nm (IQR −28 to +7)